Literature Status: Quasi-Greedy Bases in ℓ_p, $0 < p < 1$, Are Democratic

Automatic Math Discovery Run `fa_banach_001`

June 16, 2026

Original Problem

The source paper is Fernando Albiac, Jose L. Ansorena, and Przemyslaw Wojtaszczyk, *On certain subspaces of ℓ_p for $0 < p \leq 1$ and their applications to conditional quasi-greedy bases in p-Banach spaces*, arXiv:1912.08449; later published in *Math. Ann.* 379 (2021), no. 1–2, 465–502.

After Proposition 4.7, the paper remarks that the known examples of quasi-greedy bases in ℓ_p, $0 < p < 1$, are democratic, and asks:

Is every quasi-greedy basis in ℓ_p for $0 < p < 1$ democratic?

In the parsed local source this appears at `data/parsed/arxiv_sources/1912.08449/source.tex`, lines 731–735.

Answer Source

The direct answer source is Fernando Albiac, Jose L. Ansorena, and Przemyslaw Wojtaszczyk, *Quasi-greedy bases in the spaces ℓ_p $(0 < p < 1)$ are democratic*, arXiv:2004.05206; later published in *J. Funct. Anal.* 280 (2021), no. 7, 108871.

Its abstract states that the paper proves all quasi-greedy bases in ℓ_p, $0 < p < 1$, are democratic and have fundamental function of the same order as $m^{1/p}$. The main theorem makes the statement precise: if $\mathcal{X}$ is a quasi-greedy basis of ℓ_p, $0 < p < 1$, then

$$\varphi_l[\mathcal{X}, \ell_p](m) \approx m^{1/p} \approx \varphi_u[\mathcal{X}, \ell_p](m), \quad m \in \mathbb{N}.$$

In particular, $\mathcal{X}$ is democratic. In the parsed local source these are lines 125 and 266–271 of `data/parsed/arxiv_sources/2004.05206/source.tex`.

The introduction of arXiv:2004.05206 also points back to arXiv:1912.08449 as the recent paper whose examples motivated the conjectural pattern; see line 264 and the bibliography entry at lines 781–788.

Corroborating Later Source

The later paper Fernando Albiac, Jose L. Ansorena, and Glenier Bello, *Democracy of quasi-greedy bases in p-Banach spaces with applications to the efficiency of the TGA in the Hardy spaces $H_p(\mathbb{D}^d)$*, arXiv:2208.09342, cites the 2021 JFA article by title in its abstract and bibliography. This paper uses the ell$_p$ result as established literature while solving a further Hardy-space question raised in that 2021 work.

Conclusion

The question from arXiv:1912.08449 is already answered in the later literature. The answer is positive:

$$\boxed{\text{Every quasi-greedy basis in } \ell_p,\ 0 < p < 1,\ \text{is democratic.}}$$

Search Bounds

Before making this packet, the run registry, solution index, attempt index, and proof-gap index were searched for arXiv:1912.08449, arXiv:2004.05206, arXiv:2208.09342, and quasi-greedy democracy terms. No packet for this exact source question was found. A nearby existing packet records the distinct Hardy-space Question 3.8 from arXiv:2004.05206, answered by arXiv:2208.09342; this packet is only for the earlier ℓ_p question in arXiv:1912.08449.

Limitations

This packet records an already-known literature answer, not an original proof produced by the run. It does not claim any resolution of the separate q-envelope question for $\mathcal{L}_p$-spaces that appears later in arXiv:1912.08449.

Human Review Recommendation

Check that the source question on lines 731–735 of arXiv:1912.08449 is the same ℓ_p, $0 < p < 1$, democracy question, and that Theorem 1 of arXiv:2004.05206 gives the stated positive answer.

References

[1] F. Albiac, J. L. Ansorena, and P. Wojtaszczyk, *On certain subspaces of ℓ_p for $0 < p \leq 1$ and their applications to conditional quasi-greedy bases in p-Banach spaces*, Math. Ann. 379 (2021), no. 1–2, 465–502; arXiv:1912.08449.

[2] F. Albiac, J. L. Ansorena, and P. Wojtaszczyk, *Quasi-greedy bases in the spaces ℓ_p $(0 < p < 1)$ are democratic*, J. Funct. Anal. 280 (2021), no. 7, 108871; arXiv:2004.05206.

[3] F. Albiac, J. L. Ansorena, and G. Bello, *Democracy of quasi-greedy bases in p-Banach spaces with applications to the efficiency of the TGA in the Hardy spaces $H_p(\mathbb{D}^d)$*, arXiv:2208.09342.